

Farina Hussain

United States

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<https://github.com/farinah26/Resume-projects>

SUMMARY

Solutions Engineer with experience building practical systems that combine cloud services, data pipelines, and modern automation techniques to solve real business problems. I've worked on several end-to-end projects—from serverless applications to feedback analysis platforms—using tools like Lambda, DynamoDB, API Gateway, Glue, and Comprehend. My background at American Airlines and Southern California Edison has given me hands-on experience improving system reliability, streamlining workflows, and supporting decisions with clear data insights. I enjoy understanding how a problem fits into the bigger picture and designing solutions that are reliable, scalable, and simple to maintain.

EDUCATION

California State University of Los Angeles
BSc, Computer Science

Sep 2019 - Jun 2023

SKILLS

- **Cloud & DevOps:** AWS (Lambda, API Gateway, DynamoDB, S3, GlueETL, IAM), CI/CD Methodologies, Linux
- **Programming & Data:** Java, SQL, REST APIs, NoSQL (DynamoDB), ETL Pipelines
- **AI:** Amazon Bedrock (Claude), Prompt Engineering, Amazon Comprehend
- **Automation & Testing:** Selenium, TestNG
- **Analytics:** Power BI, Tableau 10, ArcGIS Pro
- **Tools:** Git, Jira, Jupyter Notebook, Agile Methodologies, AWS, Kanban, Wrike, VMWare (Basic)
- **Soft Skills:** Problem-solving, Critical Thinking, Innovation, Teamwork, Detail Oriented

PROJECTS

AI Admissions FAQ Chatbot | [Github Link](#)

Oct 2025 - Nov 2025

Independent Project

Los Angeles

- Designed and delivered a fully serverless admissions FAQ application using AWS Lambda (Java), API Gateway, DynamoDB, and S3, demonstrating practical experience building and deploying scalable, cost-efficient cloud solutions end-to-end.
- Connected a Java backend with DynamoDB to assemble relevant FAQ data and routed it through Amazon Bedrock for generated responses, implementing structured request handling and reliable prompt-based output formatting.
- Managed deployment architecture including IAM roles, environment variables, CORS configuration, and CloudWatch logging to ensure secure communication between services and stable production-style performance.

Customer Feedback Analysis | [Github Link](#)

Nov 2025 - Dec 2025

Independent Project

Los Angeles

- Developed two versions of a customer feedback analysis system, one using Python and another using Java, to evaluate design trade-offs in performance, maintainability, and integration patterns across AWS services.
- Built an automated pipeline using S3, Glue ETL, and DynamoDB to structure unprocessed customer comments, applying Amazon Comprehend for sentiment and key-pharse extraction and comparing language SDK behavior and throughput across both implementations.
- Created interactive dashboards in Amazon QuickSight to visualize sentiment trends and recurring issues, enabling data-driven improvements and demonstrating full lifecycle delivery from raw ingestion to business-level insights.

Amazon Lex Chat Bot

Jul 2022 - Aug 2022

California State University of Los Angeles

Los Angeles, CA

- Built a chatbot using Amazon Lex and DynamoDB that responded to common banking inquiries by retrieving structured account information.

WORK EXPERIENCE

American Airlines

Apr 2021 - Present

Quality Assurance Engineer

El Segundo, CA

- Designed and optimized the architecture of NetTracer's baggage tracking system by analyzing large-scale data using SQL and Java, identifying system inefficiencies and reducing mishandling incidents by 20%.
- Collaborated cross-functionally with engineering and business teams to monitor end-to-end software performance, resolve complex data discrepancies, and implement scalable system enhancements using Java-based insights for high-volume ticket resolution workflows.
- Architected and maintained automated validation frameworks leveraging Java, Selenium, TestNG, and JUnit, ensuring robust data integrity and seamless cross-platform functionality across integrated systems.

- Streamlined issue tracking and resolution processes by enhancing Jira workflows; applied architectural best practices to improve debugging efficiency and system uptime, including manual validation using VMware and structured test plans.
- Led the design and deployment of CI/CD pipelines to support automated, version-controlled releases-enhancing software delivery speed, reducing integration errors, and aligning releases with architectural roadmaps.
- Built enterprise-grade dashboards using Tableau and Power BI, integrating custom Java-based ETL logic to visualize key performance indicators, drive architectural decision-making, and align QA metrics with business goals.
- Directed A/B testing strategies for user interface enhancements, architecting backend logic in Java to gather actionable insights, refine system usability, and improve user satisfaction across critical touchpoints.

Southern California Edison (SCE)

Jan 2023 - Jan 2024

QA Software Engineer (Contract)

Santa Ana, CA

- Architected and implemented an AI-driven DRT Viewer to intelligently tag and categorize infield equipment images, significantly improving data organization, enhancing resource planning efficiency, and accelerating project delivery timelines.
- Engineered predictive algorithms using SQL and Python to optimize equipment access paths and verify asset serial numbers across distributed environments, resulting in improved system-wide data accuracy and operational reliability.
- Led the end-to-end design and deployment of an automated Resolutions Portal, applying scalable data validation architectures to strengthen connectivity integrity and reduce data-related errors by 14% across critical infrastructure systems.
- Conducted comprehensive diagnostic analysis on asset connectivity issues, synthesizing data into actionable insights that informed architectural improvements and elevated enterprise-level QA processes.
- Partnered with cross-functional teams to design and deploy interactive dashboards in Power BI and Tableau, integrating system data pipelines to highlight inconsistencies, surface KPIs, and enable data-driven decisions for stakeholders.

California State University of Dominguez Hills, IT Department

Jun 2021 - Dec 2021

Software Engineering Intern

Carson, CA

- Designed and enhanced an AI-driven chatbot architecture using Amazon Web Services (AWS), optimizing conversational flows and backend integration—resulting in a 17% boost in user engagement and system responsiveness.
- Configured the University EMS Calendaring system using the Apache Java framework, streamlining room booking workflows and improving scheduling efficiency for campus facility operations.
- Executed comprehensive QA strategies, integrating manual and automated testing frameworks for the Cal Apply system; authored technical documentation and debugging protocols that strengthened system reliability and supported long-term maintainability of the EMS platform.

CERTIFICATION

- **Harvard's CS50: Introduction to Computer Science:**edX, 2025
- **Building Test Automation Framework using Selenium and TestNG:**Coursera, 2022
- **Java Programming: Complete Beginner to Advanced:**Udemy, 2021